

ARNOLD KUMAR M

Senior AI/ML Engineer

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 8+ years of experience building machine learning systems, data pipelines, and Generative AI solutions across network infrastructure, healthcare, and financial services. My career started in Python development at Accenture in India, moved into data science and ML engineering at Bank of America and UnitedHealth Group, and has focused on production GenAI work for the last several years. I am currently at Forward Networks building LLM-powered features within their enterprise network digital twin platform, working across AWS, Azure, and GCP. I have hands-on experience across the full ML lifecycle and have spent a significant part of my career working through the reliability, scalability, and observability challenges that come with running AI systems in production at scale.

TECHNICAL SKILLS

ML & AI Frameworks	PyTorch, Hugging Face Transformers, TensorFlow, Keras, Scikit-Learn, XGBoost, LightGBM, spaCy, Sentence Transformers, LangChain, LangGraph, LangSmith
Generative AI & NLP	LLMs, RAG, Prompt Engineering, LoRA, QLoRA, PEFT, OpenAI GPT-4/3.5, Anthropic Claude, AutoGen, Amazon Bedrock, Mistral, Pinecone, Amazon OpenSearch, Agentic AI
Data Engineering & ETL	Python, Apache Spark, Apache Kafka, Apache Airflow, AWS Glue, Azure Data Factory, Delta Lake, PostgreSQL, MySQL, Elasticsearch, PySpark, Pandas, NumPy, AWS S3
MLOps & LLMOps	AWS SageMaker, Azure Machine Learning, Databricks, MLflow, CI/CD (GitHub Actions, Jenkins, Terraform), Prometheus, Grafana
Development & Deploy	Python, FastAPI, Flask, Django, REST APIs, Streamlit, Docker, Kubernetes (AWS EKS / Azure AKS), Terraform, Git/GitHub, Jupyter
Cloud Platforms	AWS: SageMaker, S3, RDS, DynamoDB, Glue, Athena, Redshift, Lambda, EC2, Kinesis, CloudWatch Azure: ML, Blob Storage, Data Factory, Databricks, Cognitive Services, AKS, Synapse Analytics GCP: Vertex AI, BigQuery, Cloud Run
Collaboration	Jira, Confluence, Agile/Scrum, SharePoint, ServiceNow, Trello

WORK EXPERIENCE

Forward Networks | United States

Senior AI/ML Engineer -- Generative AI

Jan 2025 -- Present

Project Overview: Part of Forward Networks' AI engineering team building GenAI capabilities within the Forward Enterprise network digital twin platform. The work centers on LLM-powered natural language interfaces, agentic AI workflows, and RAG-based network intelligence for NetOps, SecOps, and CloudOps teams working across complex multi-vendor network environments. Forward Networks serves Fortune 500 enterprises and government agencies across financial services, technology, and defense sectors.

- Led development of **agentic AI workflows** using **LangChain** and **LangGraph**, enabling network and security operations teams to automate multi-step network queries, compliance checks, and troubleshooting workflows through natural language, reducing manual effort and time-to-resolution for operations teams.
- Designed and built production **RAG pipelines** with hybrid semantic and keyword retrieval using **Amazon OpenSearch** and **Pinecone**, optimizing chunking strategies and reranking to improve retrieval accuracy across large-scale network device configuration and infrastructure data.

- Built **LLM evaluation** frameworks using **LangSmith** and custom metrics covering faithfulness, context recall, and answer relevance, focused on reducing hallucination rates on network-specific queries where accuracy directly impacts operational decisions.
- Fine-tuned LLMs using **PEFT (LoRA/QLoRA)** on networking-specific datasets including device configurations, query libraries, and network behavior data, improving domain accuracy and response quality for the Forward Enterprise platform.
- Built **FastAPI** microservices exposing RAG and GenAI capabilities via a unified API layer, containerized with **Docker** on **Azure AKS**. Latency and throughput under load required ongoing profiling and infrastructure tuning to keep response times acceptable in production.
- Implemented **NeMo Guardrails** to enforce output safety and ensure AI-generated network recommendations stay grounded in verified data from the Forward Enterprise digital twin, maintaining trust in AI outputs for operations teams.
- Set up **Prometheus** and **Grafana** observability across GenAI services and built automated ingestion pipelines from **SharePoint**, **ServiceNow**, and **AWS S3**, processing large daily document volumes to keep platform data current.
- Worked closely with product and platform teams to translate network operations requirements into GenAI features, and mentored junior engineers on LLM system design, evaluation frameworks, and production reliability practices.

Key Technologies: Python, FastAPI, LangChain, LangGraph, Sentence Transformers, OpenAI (Azure), Anthropic Claude (Bedrock), Mistral, Hugging Face, LoRA, QLoRA, PEFT, Amazon OpenSearch, Pinecone, AWS S3, Jenkins, Terraform, Azure AKS, Prometheus, Grafana, LangSmith, MLflow

UnitedHealth Group | United States

Jan 2023 -- Dec 2024

ML Engineer

Project Overview: Part of Optum's ML engineering team building Azure-native solutions to automate support ticket triage and analyze patient and provider feedback at scale. The work involved large-scale healthcare data, NLP-driven classification models, anomaly detection workflows, and HIPAA-compliant data governance throughout the entire pipeline.

- Designed ETL pipelines using **Azure Data Factory** and **Azure Databricks** (PySpark) to ingest, clean, and process large volumes of healthcare support ticket data, building ML-ready datasets for downstream classification and prioritization models.
- Built an automated ticket prioritization system using regression and classification models with **Azure Cognitive Services** and custom NLP classifiers, improving resolution times for critical tickets and reducing backlog across healthcare operations teams.
- Built **anomaly detection** workflows to identify irregular patterns in patient-provider interactions and claims data, helping the team flag potential fraud cases for further investigation and review.
- Owned the full **MLOps lifecycle** on **Azure Machine Learning** with **Azure DevOps**, Docker, and AKS, responsible for model deployment, versioning, monitoring, and maintaining inference reliability across production endpoints.
- Ensured **HIPAA-compliant** data governance across all pipelines using **Azure Blob Storage**, role-based access controls, and automated compliance checks, maintaining audit-readiness across all healthcare data workflows.
- Built **Power BI** dashboards to surface sentiment trends, ticket resolution KPIs, and SLA compliance metrics for healthcare operations and leadership stakeholders.
- Used **GCP Vertex AI** and **BigQuery** for supplementary model experimentation and large-scale data querying, supporting multi-cloud MLOps requirements across the Optum platform.

Key Technologies: Python, SQL, PySpark, Scikit-Learn, PyTorch, spaCy, Azure Machine Learning, Azure Cognitive Services, Azure Databricks, Azure Data Factory, Azure Blob Storage, Azure Synapse Analytics, Docker, AKS, Azure Functions, GitHub, Azure DevOps, Terraform, Power BI, Vertex AI, BigQuery

Bank of America | United States

Oct 2019 -- Dec 2022

Data Scientist

Project Overview: Led development of a credit risk analytics platform to predict credit card defaults and guide lending decisions. The work involved handling large-scale imbalanced financial data, building and tuning ML models, and transitioning the team from a nightly batch scoring process to near-real-time risk scoring through streaming data pipelines.

- Built **credit risk classification models** using **XGBoost** to identify high-risk customers, improving default prediction accuracy and reducing missed-risk cases for the lending team, which had a direct impact on credit loss management.

- Addressed **class imbalance** using **SMOTE** and threshold optimization to improve high-risk recall in real-world lending scenarios, where false negatives carried significant business cost and needed to be minimized.
- Developed scalable data pipelines using **Python** and **PySpark** to process large financial transaction volumes, significantly reducing nightly batch processing times and improving data freshness for downstream model scoring.
- Implemented real-time data ingestion using **Apache Kafka** and **AWS Kinesis**, enabling near real-time risk scoring that supported faster lending decisions and reduced the lag between transaction and risk assessment.
- Built **Tableau** dashboards and automated operational workflows with **Apache Airflow** and Docker, improving stakeholder visibility into model performance, deployment health, and pipeline status.
- Worked closely with the credit risk team to evaluate models using ROC-AUC, precision-recall curves, and threshold analysis, calibrating outputs to align with lending policy requirements and business risk tolerance.

Key Technologies: Python, PySpark, Apache Kafka, AWS Kinesis, AWS S3, AWS Glue, Amazon Redshift, Docker, PostgreSQL, Apache Airflow, Elasticsearch, Tableau, Git, REST APIs, CI/CD

Accenture | India

May 2017 -- Sep 2019

Python Developer

Project Overview: Worked across retail, energy, and telecom clients building Python automation scripts, RESTful APIs, and data visualization tooling. This role provided hands-on experience delivering production software across multiple client environments and built the engineering foundation for the data science and ML work that followed.

- Built **RESTful APIs** with **Flask** to deliver KPIs and operational data in real time for business teams, significantly reducing manual reporting turnaround and enabling faster data-driven decisions.
- Built **Python ETL pipelines** using Pandas and **SQLAlchemy** to process weekly customer records across multiple client datasets, reducing manual workload and eliminating several recurring reporting tasks that had previously required dedicated headcount.
- Developed **Tableau** dashboards and optimized SQL queries to enable self-service analytics for non-technical stakeholders across retail, telecom, and energy divisions, reducing dependency on engineering teams for routine reporting.
- Wrote **PyTest** unit and integration test suites and built reusable Python libraries shared across the team, improving code reliability and reducing bug rate across multiple client projects.
- Designed **PostgreSQL** and **MySQL** database schemas supporting BI reporting and operational workflows across retail, telecom, and energy business units.

Key Technologies: Python, Flask, Django, Pandas, SQLAlchemy, PostgreSQL, MySQL, Tableau, Matplotlib, Seaborn, PyTest, Linux scripting, Git, REST APIs, CI/CD, Jenkins

NOTABLE PROJECTS

Network Operations RAG Agent *LangChain · LangGraph · Pinecone · Amazon OpenSearch · FastAPI · NeMo Guardrails*

Designed and deployed a production RAG system within Forward Networks to enable natural language querying of network device configurations and infrastructure data. Used multi-agent orchestration with hybrid semantic-keyword retrieval, LLM guardrail pipelines, and LangSmith-based evaluation to improve retrieval accuracy and reduce time-to-resolution for NetOps and SecOps teams. A key challenge was keeping responses grounded in verified network state data rather than hallucinated outputs.

Healthcare Ticket Triage and Fraud Analytics Platform *Azure ML · Azure Databricks · PySpark · Azure Cognitive Services · spaCy · Docker · Power BI*

Built an Azure-native MLOps platform at UnitedHealth Group to automate support ticket prioritization and detect anomalies in claims data. Combined NLP-driven multi-class classification with anomaly detection models, deployed on Azure Kubernetes Service with full HIPAA-compliant data governance. The ticket triage system reduced response times for critical healthcare operations tickets, and the anomaly detection workflow helped flag potential fraud cases for the compliance team.

Real-Time Credit Risk Scoring Engine *XGBoost · SMOTE · Apache Kafka · AWS Kinesis · PySpark · Amazon Redshift · Apache Airflow*

Led the transition from nightly batch credit risk scoring to a near-real-time streaming platform at Bank of America. An XGBoost classifier trained on SMOTE-balanced transaction data was integrated into Kafka and AWS Kinesis ingestion pipelines, enabling same-session lending decisions. Getting class imbalance right was the most challenging part of the modeling work, and a significant amount of time went into threshold tuning to align model outputs with the credit team's risk tolerance.

CERTIFICATIONS

- AWS Certified Machine Learning -- Specialty
- Microsoft Certified: Azure AI Engineer Associate
- GCP Professional Machine Learning Engineer
- Generative AI with LLMs -- DeepLearning.AI

EDUCATION

Master of Science, Artificial Intelligence Engineering

University of North Texas | Denton, Texas, USA